

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_12apr25_ajust.parquet

model_type: GRU
group_sec: 40
timesteps_orig: 60
batch_size: 256
min_units: 32
max_units: 256
units_step: 32
dropout_min: 0.2
dropout_max: 0.3
dropout_step: 0.1
activation_functions: ['tanh', 'relu', 'elu']
optimizers: ['adam', 'rmsprop', 'Nadam']
min_layers: 1
max_layers: 2
max_epochs: 150
hyperband_iterations: 3
group_msec: 40
samples_per_sec: 25
forecast_horizon_sec: 8
forecast_steps: 200
timesteps: 60

GRU Multi-Step Training Report

Report Generated: 2025-05-17 07:45:46

Training Start: 2025-05-16 22-00-39

Training End: 2025-05-16 22-24-03

Training Duration: 23.40 minutes

Predictions Start: 2025-05-16 22-24-03

Predictions End: 2025-05-16 22-27-09

Predictions Duration: 3.10 minutes

Best Hyperparameters:

Number of Layers: 2

Activation: tanh

Optimizer: rmsprop

Dropout Rate: 0.20

MAE: 2.1274

MSE: 6.5982

RMSE: 2.5687

R2 Score: 0.6086

Multi-Step Forecast:

Forecast Steps: 200

MAE: 0.0359

MSE: 0.0019

RMSE: 0.0436

R2 Score: 0.8013

Dataset size before down-sample: 85,269 rows
Dataset size after down-sample: 12,829 rows

First 5 timestamps BEFORE:

1744397085760
1744397085770
1744397085789
1744397085801
1744397085817

...

Last 5 timestamps BEFORE:

1744464971366
1744464971377
1744464971390
1744464971402
1744464971414

First 5 timestamps AFTER:

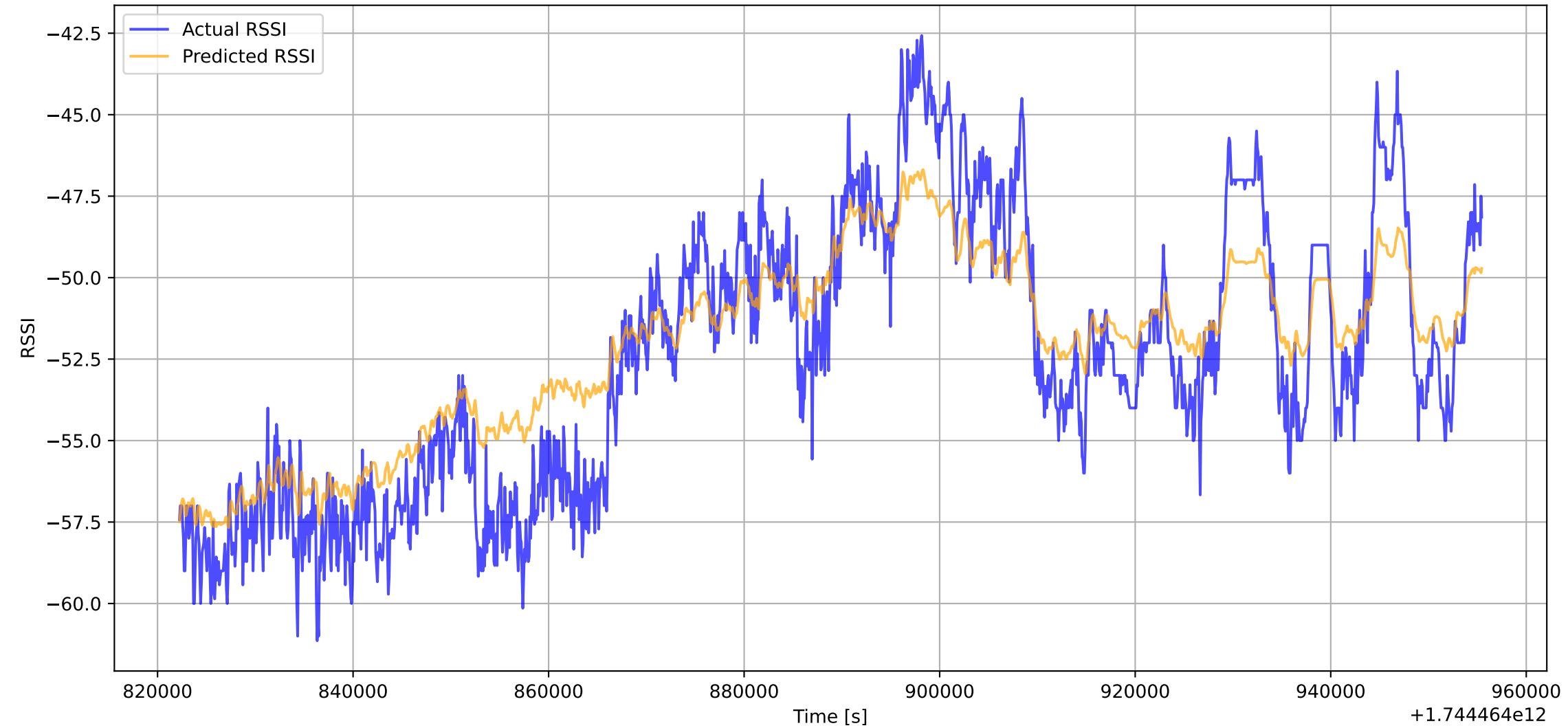
1744397085760
1744397085841
1744397086014
1744397086103
1744397086172

...

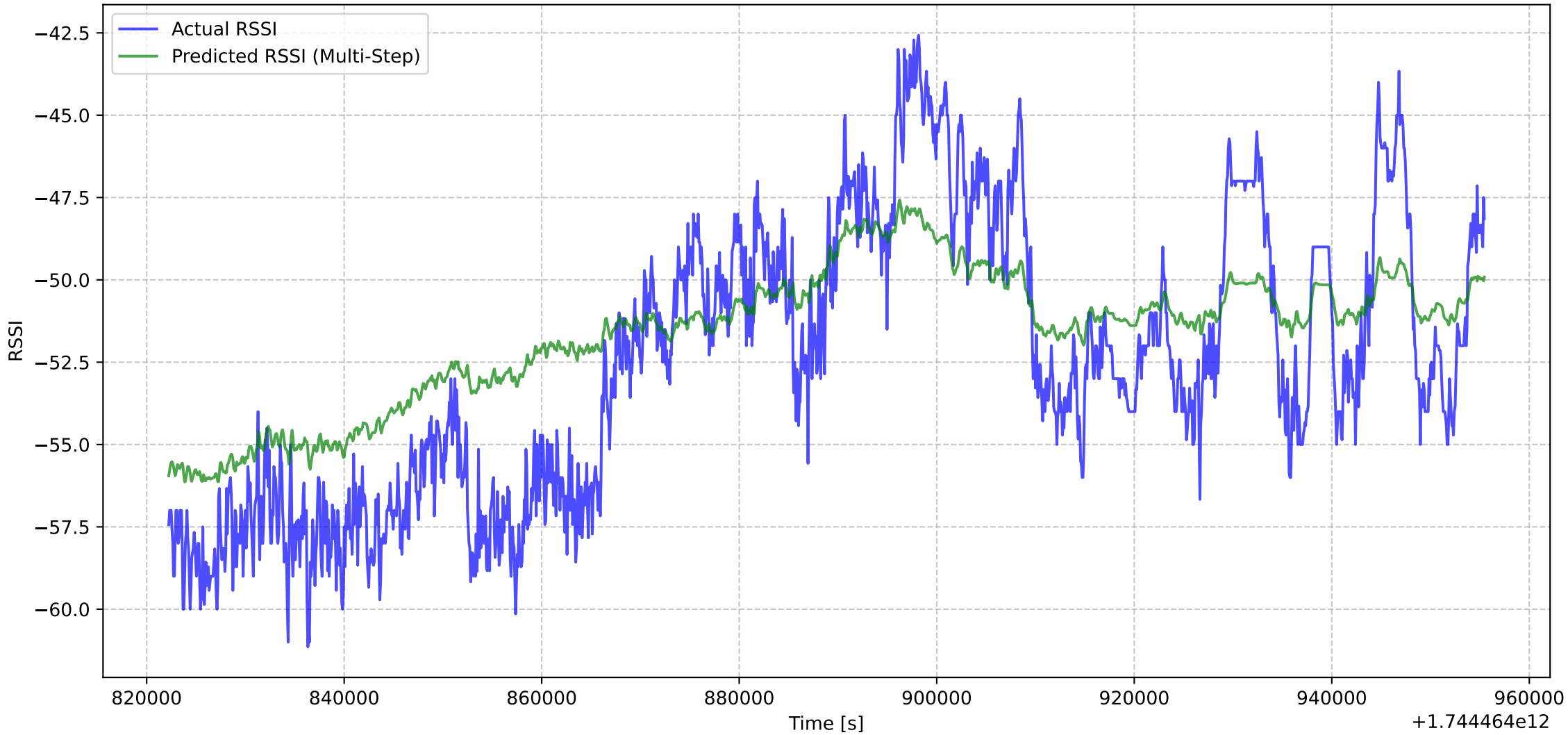
Last 5 timestamps AFTER:

1744464971041
1744464971124
1744464971209
1744464971281
1744464971366

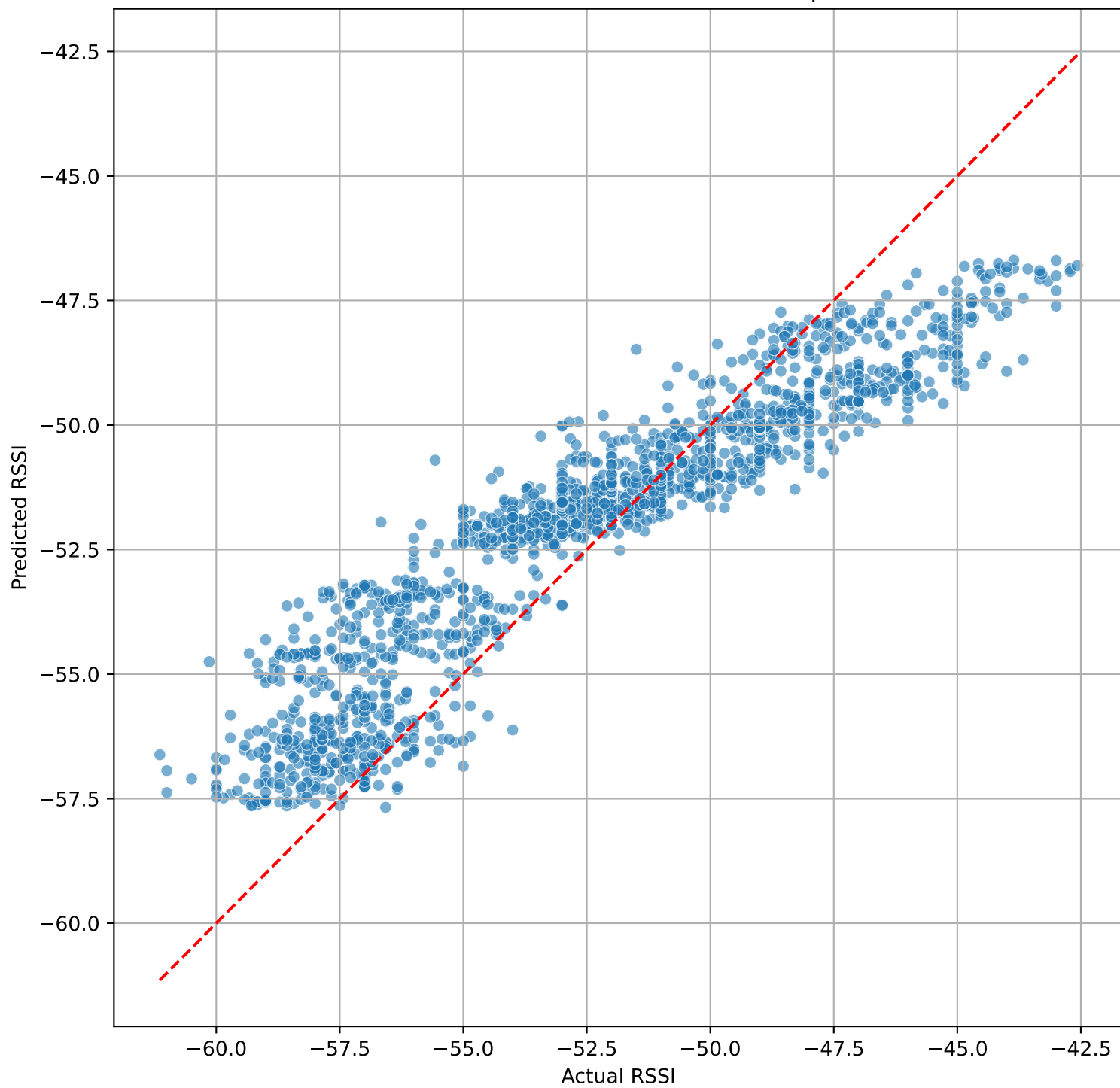
Actual vs Predicted RSSI (First Step)



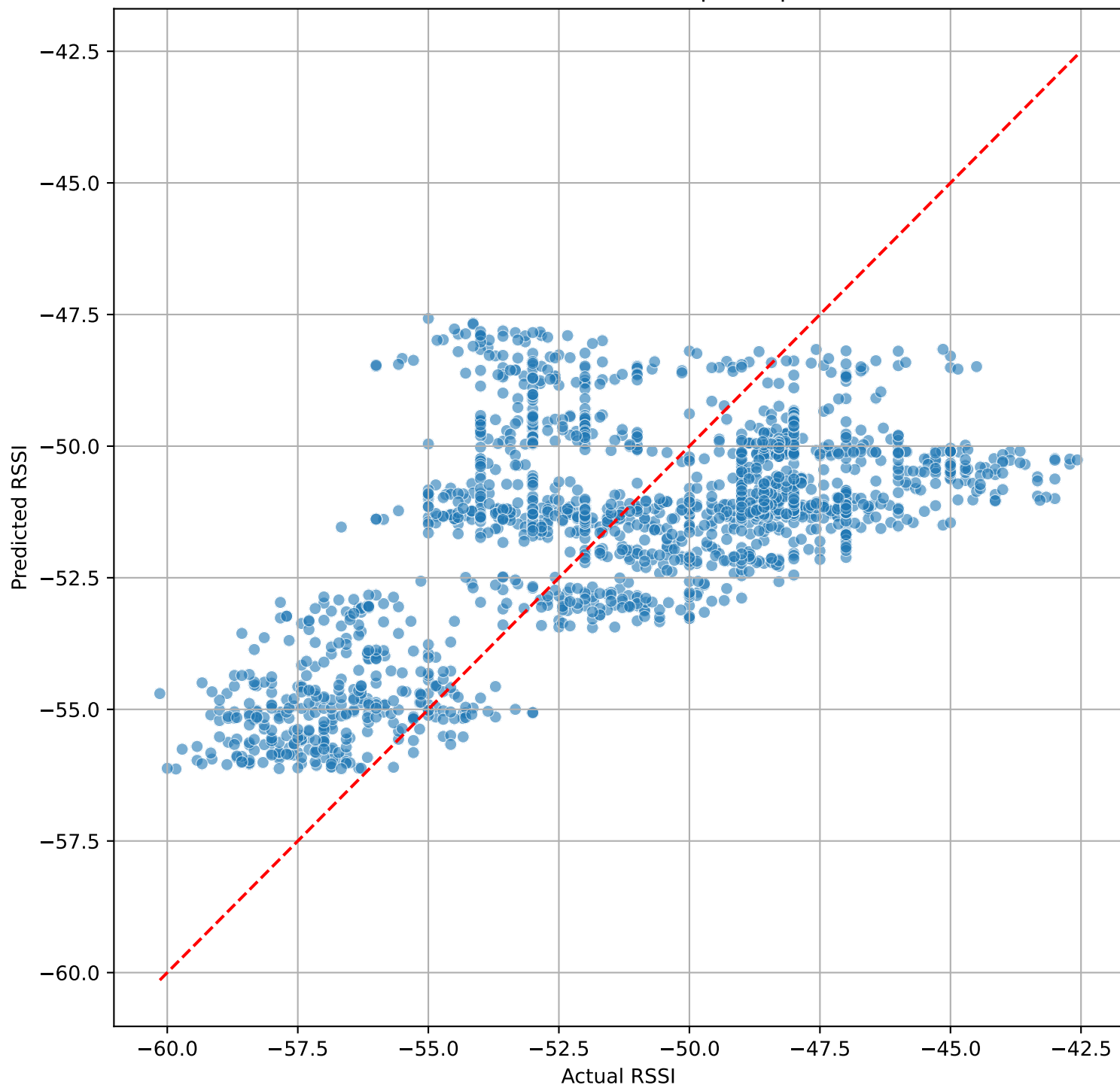
Actual vs Predicted RSSI (Multi-Step (step 200 $\approx$ 8.0 s))



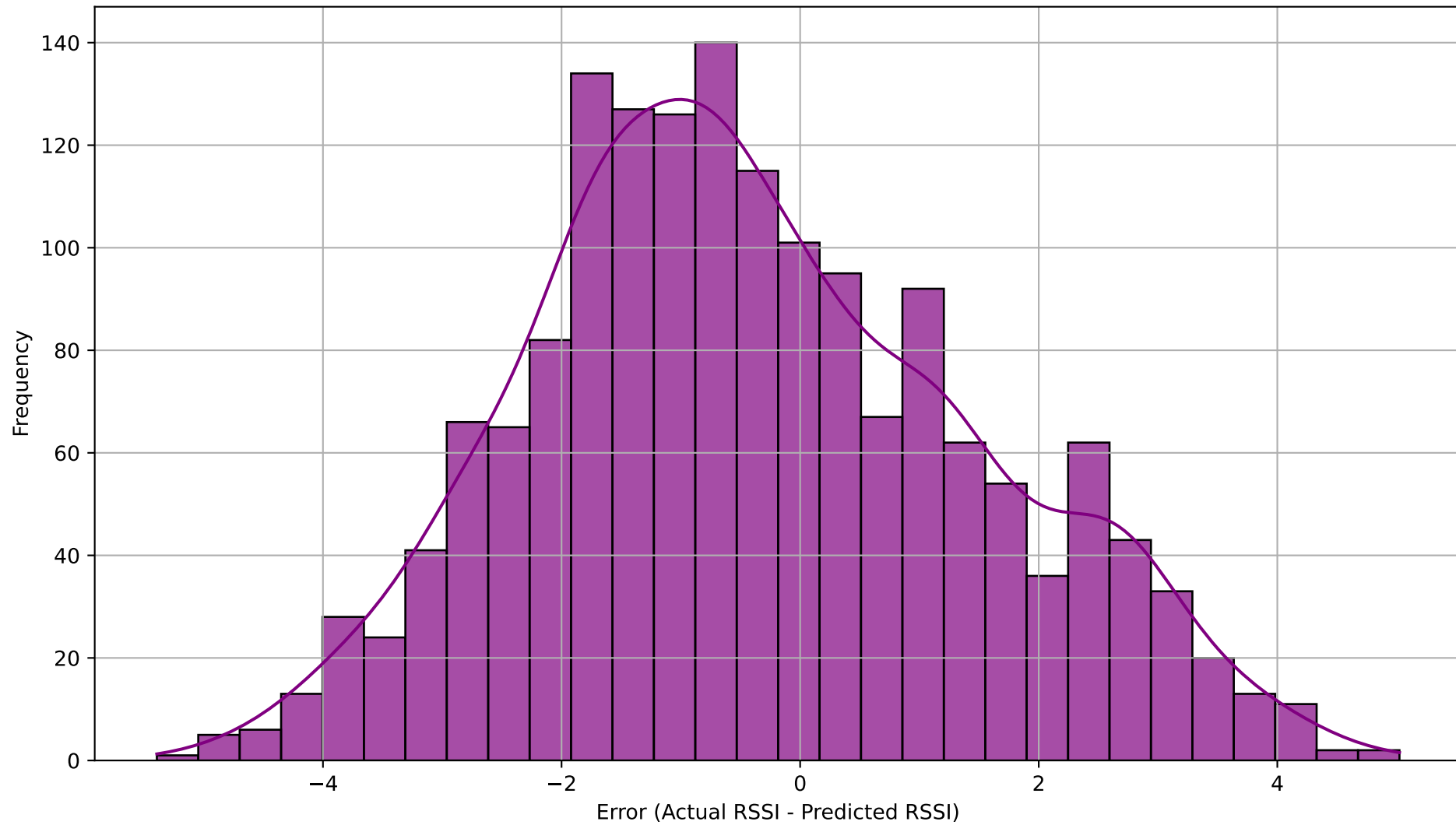
Actual vs Predicted RSSI (First Step)



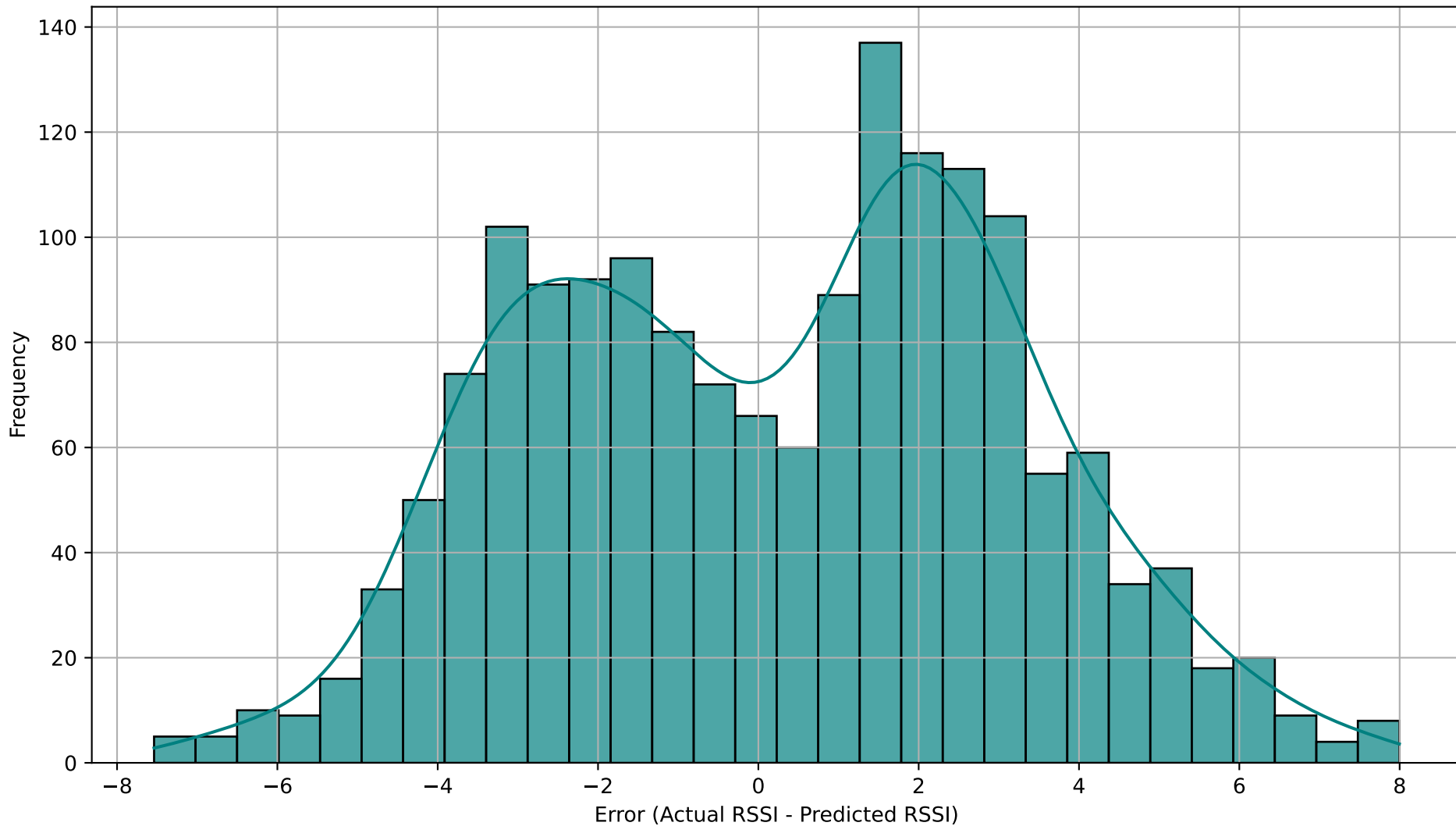
Actual vs Predicted RSSI (Last Step - step 200 $\approx$ 8.0 s)



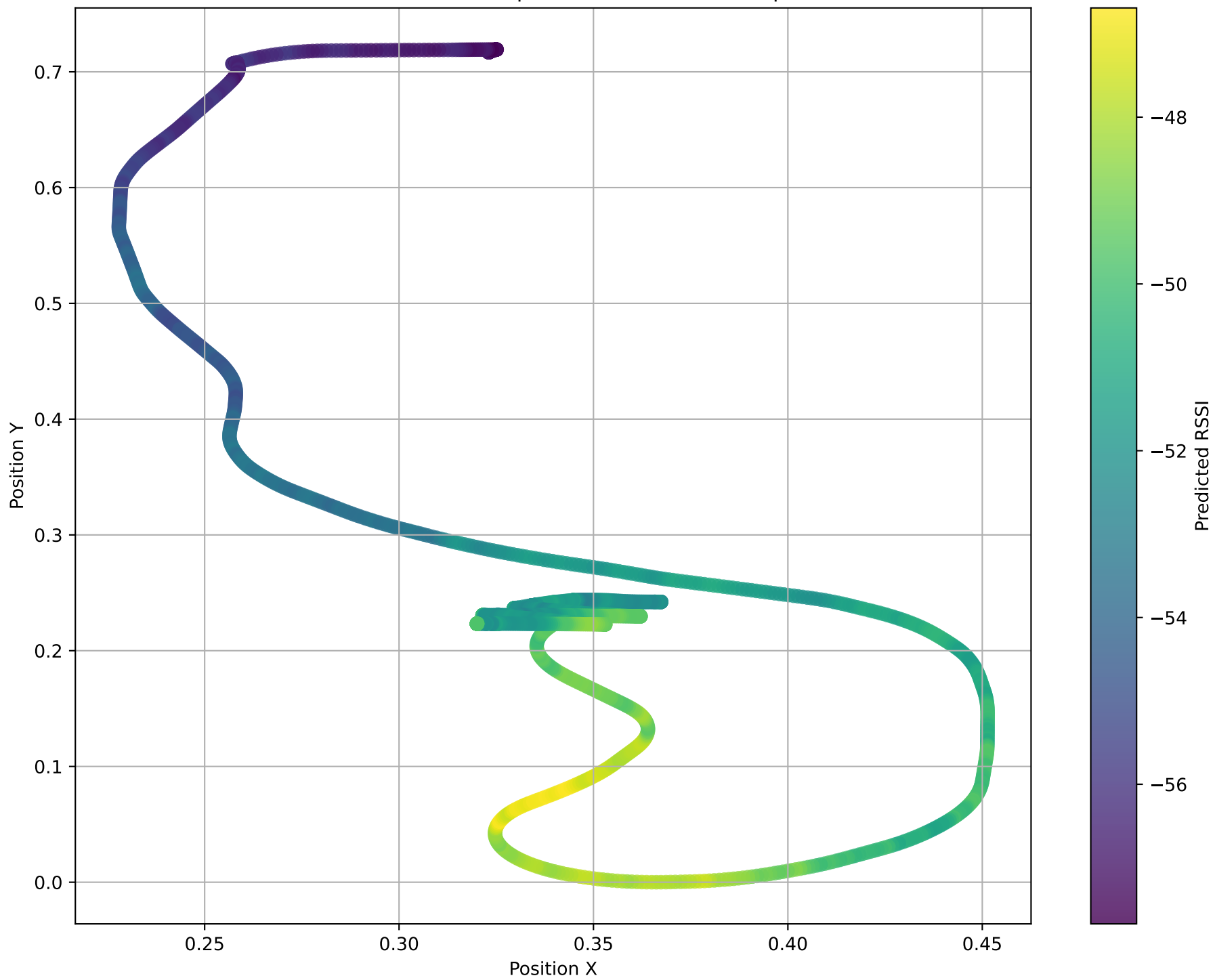
Distribution of Prediction Errors (First Step)



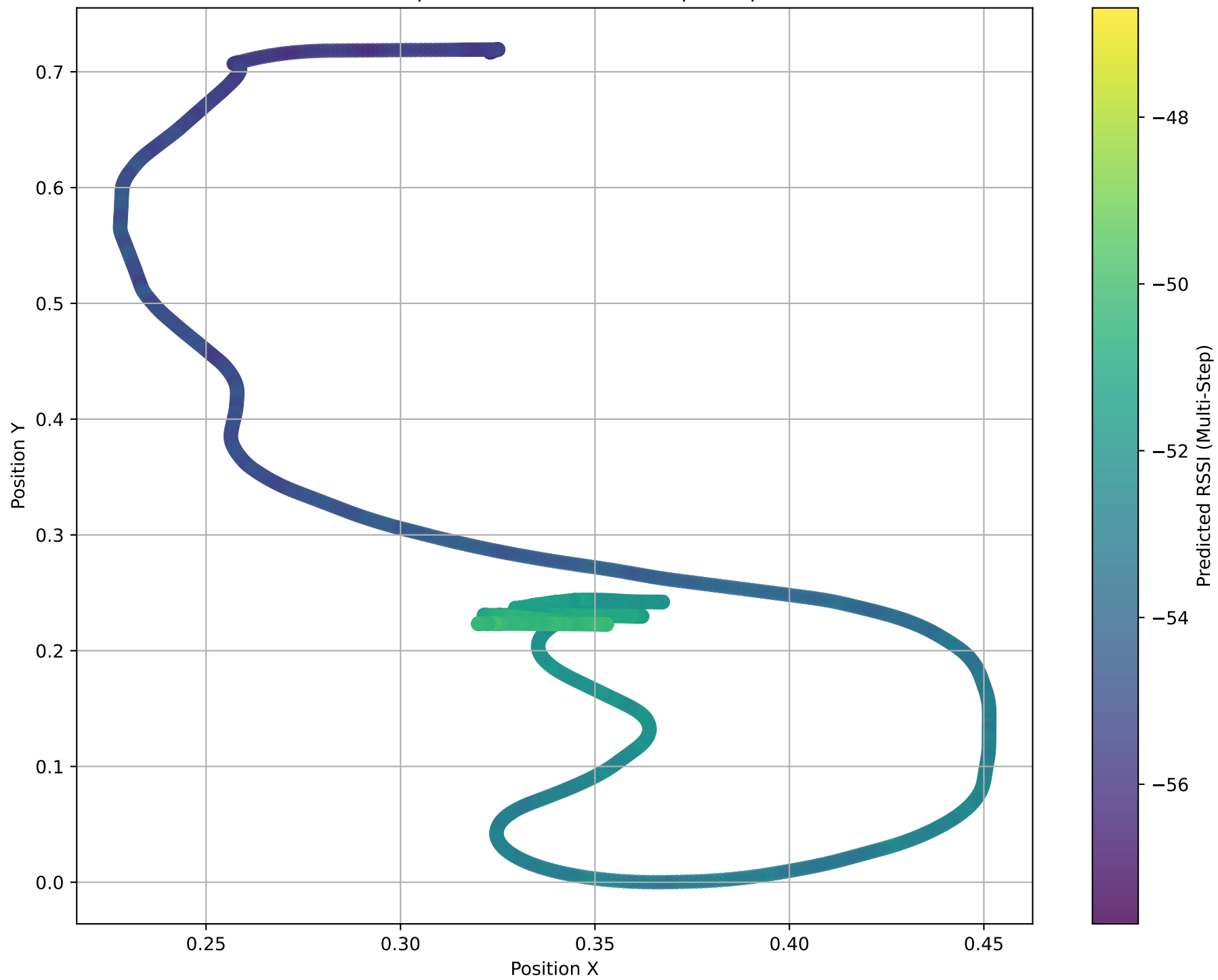
Distribution of Prediction Errors (Last Step - step 200 $\approx$ 8.0 s)



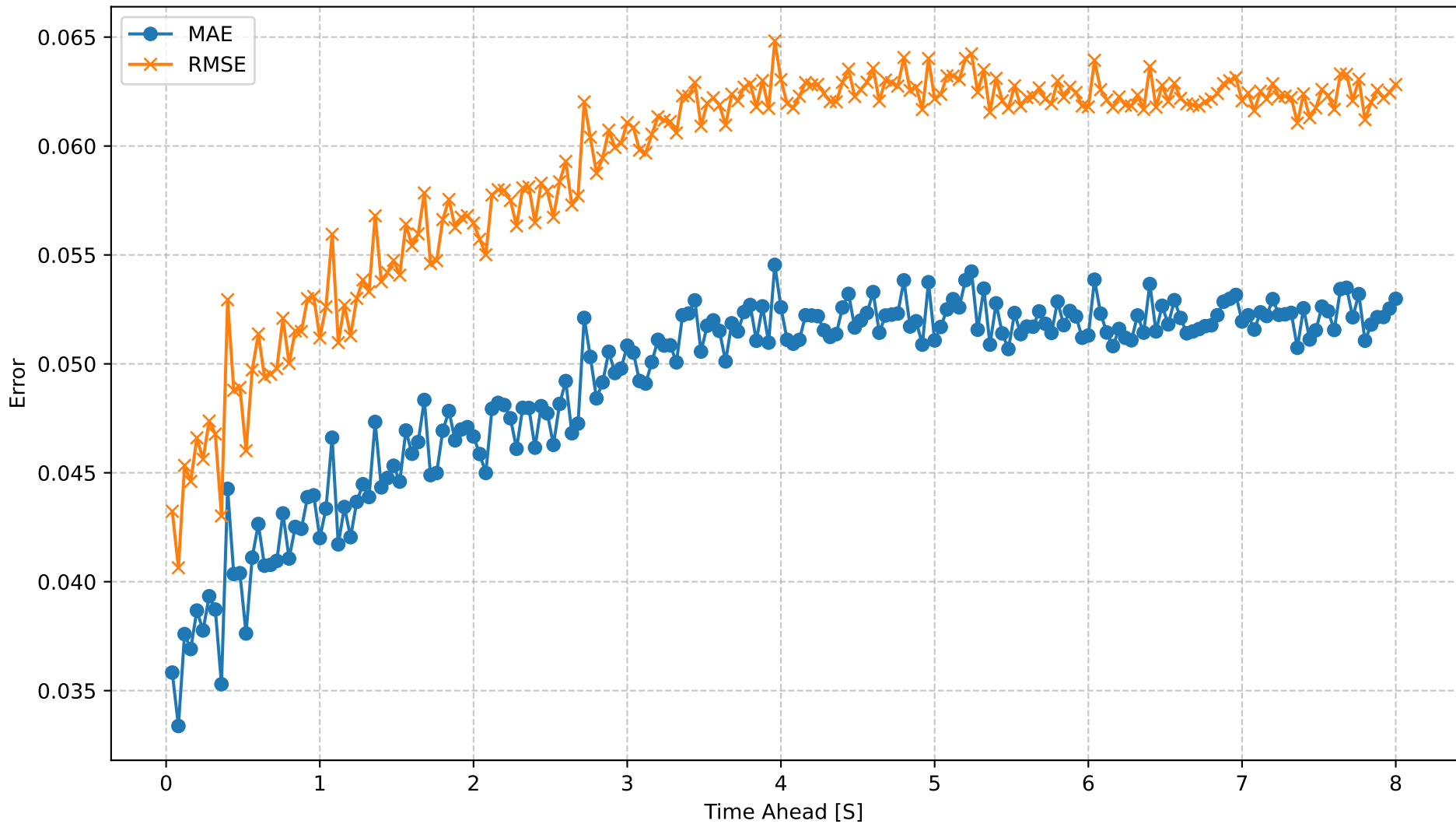
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 200 $\approx$ 8.0 s)



Multi-Step Forecast Error by Time Ahead



Resumo Automático

Dataset: rssi_odom_dataset_12apr25_ajust.parquet

Modelo: GRU

Forecast steps: 200

MAE / RMSE (dB)

- 1º passo : 0.04 / 0.04
- Mediana : 0.05 / 0.06
- 200º passo : 0.05 / 0.06

Crescimento do erro: 47.9% → Moderado (erro cresce 20-50 %)

3 piores passos (MAE): 99 (0.05), 131 (0.05), 151 (0.05)

Recomendação:

Use as previsões até a metade do horizonte; após isso a incerteza cresce consideravelmente.